

**Inclusive Educational Software System: Emotionally Expressive Reading for
Children with Disabilities (Blind, Deaf, Cognitive Impairments)**

Project ID: 25-26J-407

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specializing in Information
Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

**Inclusive Educational Software System: Emotionally Expressive Reading for
Children with Disabilities (Blind, Deaf, Cognitive Impairments)**

Emotion-Driven Story Recommendation System

Vikesh R – IT22103222

B.Sc. (Hons) Degree in Information Technology Specializing in Information
Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

ABSTRACT

Children with disabilities often face significant challenges in accessing personalized and emotionally supportive storytelling experiences, which are crucial for their emotional development and overall well-being. This research proposes an innovative Emotion-Driven Story Recommendation System tailored specifically for children with disabilities. The system integrates accessible, multimodal emotion detection techniques—leveraging emoji-based self-reporting, advanced text sentiment analysis, and optional voice emotion recognition—to accurately capture the user's emotional state. By tagging story content sentence-by-sentence based on emotional tone, the platform dynamically aligns the narrative flow with the child's current mood, fostering personalized and engaging storytelling experiences.

Unlike complex facial recognition systems that often impede accessibility, this approach ensures inclusivity by employing simple, adaptive input methods suitable for children with varying cognitive and sensory abilities. The use of multimodal emotion detection reflects contemporary findings in affective computing, which demonstrate that combining multiple channels (text, voice, and user input) enhances recognition accuracy and robustness, especially for neurodivergent children who may express emotions differently. This system's real-time mood-to-story matching aims to increase emotional engagement, improve mood regulation, and support therapeutic outcomes.

The proposed solution emphasizes cultural sensitivity and scalable implementation, with potential applicability in diverse educational and therapeutic contexts. The study will rigorously evaluate the system's performance in terms of emotion detection accuracy, recommendation relevance, and impacts on emotional well-being. By bridging gaps in assistive technology, emotion recognition research, and personalized content delivery, this work presents a novel, accessible framework for adaptive storytelling interventions designed to empower children with disabilities.

Keywords: emotion-driven recommendation, multimodal emotion detection, personalized storytelling, children with disabilities, assistive technology, accessibility, affective computing, mood-based content delivery

TABLE OF CONTENTS

Abstract	3
1. Introduction	9
1.1 Background & Literature Survey	10
1.2 Research Gap.....	12
1.3 Research Problem	14
2. Objectives.....	15
2.1 Main Objective	15
3. Methodology	17
3.1 Key Pillars And Application Of Technologies	17
3.2 Proposed Approach And Task Breakdown	18
3.4 Proposed Project Plan.....	22
3.5 Anticipated Conclusion.....	24
3.6 Commercialization.....	25
4. Project Requirements	25
4.1 Functional Requirements	25
4.2 User Requirements.....	26
4.3 System Requirements	26
4.4 Non-Functional Requirements.....	27
5. Budget And Budget Justification.....	28
6. References	29

LIST OF FIGURES

Figure 1: Component Diagram.....	19
Figure 2: Gantt Chart	23
Figure 3: Work Breakdown Structure.....	23

LIST OF TABLES

Table 1: Research Gap for component	13
Table 2: Budget Distribution	29

DECLARATION

I hereby declare that the work presented in this report of proposal is my own and has not been submitted elsewhere. All sources of information have been duly acknowledged. This report is submitted in partial fulfillment of the requirements of the above degree program, and it reflects the research and development plan for my individual component of the project.

Name	Student ID	Signature
Vikesh R	IT22103222	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



27/08/2025

.....
Signature of the Supervisor

.....
Date

Ms. Sanjeevi Chandrasiri



27/08/2025

.....
Signature of the Co-Supervisor

.....
Date

Ms. Karthiga Rajendran

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
ASD	Autism Spectrum Disorder
NLP	Natural Language Processing
ML	Machine Learning
GUI	Graphical User Interface

1. INTRODUCTION

Children with disabilities constitute a significant and vulnerable population that encounters numerous challenges in accessing engaging and developmentally appropriate educational and therapeutic resources. Among these, storytelling plays a pivotal role as a tool for emotional development, social learning, and cognitive stimulation. However, traditional storytelling methods and current digital platforms often lack adaptability to the unique emotional needs and accessibility requirements of children with diverse disabilities.

Recent advancements in affective computing and artificial intelligence have paved the way for emotion recognition technologies that can interpret human emotions through various modalities such as facial expressions, voice tone, text sentiment, and physiological signals. Integrating these technologies into interactive storytelling platforms offers a promising avenue for personalized content delivery that aligns with the child's emotional state, thereby enhancing engagement, comprehension, and emotional well-being.

Despite growing interest, existing systems predominantly rely on complex modalities like facial recognition, which may not be suitable for all children due to privacy concerns, cognitive load, or physical limitations. Additionally, few systems provide the granularity of emotion detection needed to tailor story content at the sentence level, nor do they incorporate accessible input methods to ensure usability across varying disabilities.

This research proposes an innovative Emotion-Driven Story Recommendation System specifically designed for children with disabilities. By employing multimodal emotion detection techniques—ranging from emoji-based self-reporting and text sentiment analysis to optional voice emotion recognition—and avoiding reliance on invasive or complex technologies, the system aims to offer culturally sensitive, accessible, and emotionally resonant storytelling experiences. The system dynamically tags story content based on emotional tone and matches it to the child's current mood, creating personalized narrative pathways that support emotional engagement and well-being.

The proposed approach bridges gaps at the intersection of assistive technology, affective computing, and personalized learning, promising significant implications for therapeutic interventions and inclusive education. By focusing on scalability and accessibility, this system seeks to empower children with disabilities globally, providing them with adaptive storytelling tools that nurture emotional development in a supportive environment.

1.1 Background & Literature Survey

Children with disabilities, including those with cognitive, sensory, and developmental challenges, require specialized approaches for effective learning and emotional development. Storytelling has long been recognized as a powerful pedagogical and therapeutic tool that fosters emotional literacy, empathy, and cognitive skills in children [5], [6], [2]. Research conducted in Sri Lanka highlights the impact of **non-linear storytelling**, showing that interactive narrative approaches can significantly improve social and emotional development in preschool children compared to traditional methods [2].

However, traditional storytelling approaches often lack adaptability to the individual emotional states and accessibility needs of children with disabilities, limiting their effectiveness [1], [10].

In recent years, affective computing and emotion recognition technologies have demonstrated potential in personalizing content through real-time emotional state detection using modalities such as facial expressions, voice tone, physiological signals, and text sentiment analysis [3], [7], [8]. A Sri Lankan study developed a **digital storytelling platform with sentence-level emotional tagging**, using OCR, contextual emotion recognition (CER), and emotional TTS. The system achieved OCR accuracy of 75%, CER accuracy of 72%, and Mean Opinion Scores of 3.85 for neutral TTS and 1.65 for emotional TTS—showcasing the feasibility of emotion-driven storytelling in local contexts [1].

Multimodal emotion detection systems that combine diverse inputs have been shown to improve robustness and accuracy [8], [9]. In another Sri Lankan initiative, researchers designed **Tellie**, a mobile app for vision- and hearing-impaired children. It integrates OCR, BERT-based sentence-level emotion recognition, emotionally styled TTS, and sign language conversion, offering inclusive and multimodal storytelling [3].

Several studies globally have explored integrating emotion recognition into content recommendation. Emotion-based music and movie recommender systems adapt suggestions to user moods, enhancing engagement and satisfaction [9], [10]. Yet, limited work has been done in adapting such frameworks to storytelling for children with disabilities [4].

A recurring concern in existing systems is the reliance on camera-based emotion detection, which can raise privacy issues and exclude children with physical or sensory impairments [7]. Alternative inclusive approaches—such as emoji-based self-reporting, voice emotion recognition, and text sentiment analysis—have been suggested as effective solutions [2], [3].

Despite advances both globally and in Sri Lanka, a gap remains in developing **accessible, multimodal emotion-driven storytelling systems** tailored specifically for children with disabilities. This research aims to address that gap by designing a system that combines sentence-level emotional tagging of story content with dynamic matching to user emotional states using inclusive input mechanisms [1], [2], [3], [5]. The goal is to enhance emotional engagement, support mood regulation, and provide therapeutic outcomes for diverse groups of children.

1.2 RESEARCH GAP

Despite significant advances in emotion recognition technologies and personalized recommendation systems, several gaps remain unaddressed in the context of storytelling for children with disabilities. First, while multimodal emotion detection systems have been developed to enhance accuracy by incorporating inputs such as facial expressions, voice tone, and text sentiment, limited research has focused on integrating these modalities into storytelling platforms specifically designed for children with diverse disabilities .,

Secondly, many existing emotion-aware content recommendation systems rely heavily on facial recognition, which may pose privacy, accessibility, and usability challenges for children with physical impairments or discomfort with camera-based technologies. Alternative input mechanisms, such as emoji-based self-reporting and voice-based emotion recognition, remain underexplored in personalized storytelling for this vulnerable group.

Moreover, most prior works tend to treat emotion recognition and content recommendation as separate problems, lacking a cohesive framework that seamlessly links accurate, real-time emotion detection with dynamic narrative adaptation at a granular content level such as sentence-by-sentence emotional tagging. Furthermore, cultural sensitivity and accessibility-first design approaches are often limited or omitted, reducing the applicability of such systems in diverse socio-cultural settings.

Finally, there is a scarcity of evaluation studies examining not only the technical accuracy of emotion detection but also the therapeutic and emotional well-being outcomes for children with disabilities, particularly in longitudinal or real-world settings. These gaps highlight an opportunity to develop and validate a novel Emotion-Driven Story Recommendation System that addresses these challenges by integrating multimodal emotion detection with mood-responsive, accessible storytelling content tailored to children with disabilities.

Research Gap Feature	Research 1	Research 2	Research 3	Research 4	Proposed Research Solution
Multimodal emotion detection	✓	X	✓	✓	✓
Emotion-based storytelling tailored for disabilities	X	X	X	X	✓
Accessibility without reliance on facial recognition	X	X	X	X	✓
Sentence-level emotional content tagging	X	X	X	X	✓
Culturally sensitive and scalable design	X	X	X	X	✓
Real-world evaluation on emotional well-being	X	X	X	X	✓

Table 1: Research Gap for component

1.3 RESEARCH PROBLEM

Providing personalized and emotionally responsive storytelling to children with disabilities remains a significant challenge in educational and therapeutic contexts. Although storytelling is a proven medium for fostering emotional development and social learning, existing platforms generally lack mechanisms to detect and respond to the nuanced emotional states of children, particularly those with cognitive, sensory, or communication impairments. Current emotion recognition systems often depend on single modalities such as facial expression analysis or voice recognition, which can be unreliable or inaccessible for children with different disability profiles .

Moreover, conventional story recommendation systems typically rely on user preferences or demographic information and do not incorporate real-time emotional feedback to tailor content dynamically. This gap limits the capacity to create engaging, mood-congruent stories that support emotional regulation and therapeutic goals. Few frameworks have explored the integration of multimodal emotion detection—including accessible inputs like emoji-based self-reporting, text sentiment analysis, and optional voice emotion detection—with adaptive storytelling systems.

Another substantial problem is the lack of inclusive design in existing solutions. Many depend on complex and intrusive sensing technologies such as facial recognition, which can pose privacy concerns and engineering challenges when applied to children with disabilities. Ensuring accessibility across diverse disabilities using non-invasive, adaptable input methods remains an unmet need. Consequently, the absence of a culturally sensitive, scalable system that combines multimodal emotion recognition with precise, sentence-level emotional tagging in stories prevents the realization of truly personalized and emotionally supportive storytelling experiences for children with disabilities. This research addresses this problem by proposing a novel Emotion-

Driven Story Recommendation System specifically designed with these considerations in mind, aiming to enrich children's emotional engagement and support their well-being effectively.

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate an innovative Emotion-Driven Story Recommendation System specifically tailored for children with disabilities. This system aims to integrate accessible multimodal emotion detection techniques—including emoji-based self-reporting, text sentiment analysis, and optional voice emotion recognition—with a mood-based personalized storytelling recommendation engine. By analyzing emotional tones at the sentence level within stories and dynamically matching these with the user's detected mood, the system intends to enhance emotional engagement, facilitate mood regulation, and support the overall emotional well-being of children with diverse disabilities. The system will prioritize inclusivity, scalability, and cultural sensitivity, addressing key gaps in current assistive technologies and affective computing applications.

2.2 Specific Objectives

1. Develop an accessible multimodal emotion detection framework that integrates emoji-based self-reporting, text sentiment analysis, and optional voice emotion recognition.

This framework will provide flexible and inclusive emotion input methods that cater to the varied needs of children with disabilities, avoiding intrusive technologies like facial recognition.

2. Design a sentence-level emotional tagging algorithm for storytelling content.

This algorithm will analyze and label each sentence of stories by their emotional tone, enabling precise matching of narrative segments to the user's mood.

3. Build an adaptive story recommendation engine that dynamically aligns the detected emotional state of the child with emotionally congruent story content.

This engine will tailor storytelling experiences in real time, enhancing engagement and emotional resonance.

4. Ensure the system prioritizes accessibility, inclusivity, and cultural sensitivity.

Design considerations will focus on diverse disability profiles and cultural backgrounds to maximize usability and relevance.

5. Evaluate the system's effectiveness through user studies measuring emotion detection accuracy, user engagement, and emotional well-being impacts.

Rigorous evaluation will validate the practical benefits and guide iterative improvements.

3. METHODOLOGY

This research aims to develop an emotionally expressive and user-centric Text-to-Speech (TTS) storyteller system. The methodology involves a structured approach, integrating key pillars of speech synthesis, machine learning, and mobile application development to achieve the project's main objective and specific sub-objectives. Each task and sub-task within the methodology is designed to ensure the system operates effectively, engagingly, and accessibly.

3.1 Key Pillars and Application of Technologies

- **Multimodal Emotion Detection:** Combine emoji-based self-reporting, text sentiment analysis using NLP, and voice emotion recognition via machine learning to ensure accurate, inclusive emotion capture.
- **Sentence-Level Emotional Tagging:** Apply natural language processing techniques to label emotional tone at the sentence level in stories, facilitating granular emotion-content mapping.
- **Adaptive Story Recommendation:** Develop an engine that dynamically aligns the child's current emotional state with emotionally tagged story segments for personalized storytelling.
- **Accessibility and Scalability:** Prioritize inclusive design with interfaces adaptable for various disabilities and culturally sensitive content adaptable to diverse populations.

3.2 Proposed Approach and Task Breakdown

- Emotion Data Collection: Design interactive, easy-to-use input methods (emoji, text, voice) to gather emotional states from children with diverse disabilities.
- Multimodal Emotion Detection Development: Train hybrid models integrating textual sentiment, audio emotion, and self-reported emoji data for robust mood inference.
- Story Content Emotional Analysis: Develop algorithms to analyze and tag story sentences with emotional labels through machine learning-based NLP.
- Recommendation Algorithm and System Integration: Build a recommendation system that aligns real-time detected moods with tagged story content; integrate all modules for seamless functioning.
- User Interface Design and Testing: Create accessible, child-friendly UI and conduct usability testing with target users.
- Evaluation: Collect quantitative and qualitative data on accuracy, engagement, and emotional well-being effects for iterative refinement.

3.3 System Diagram

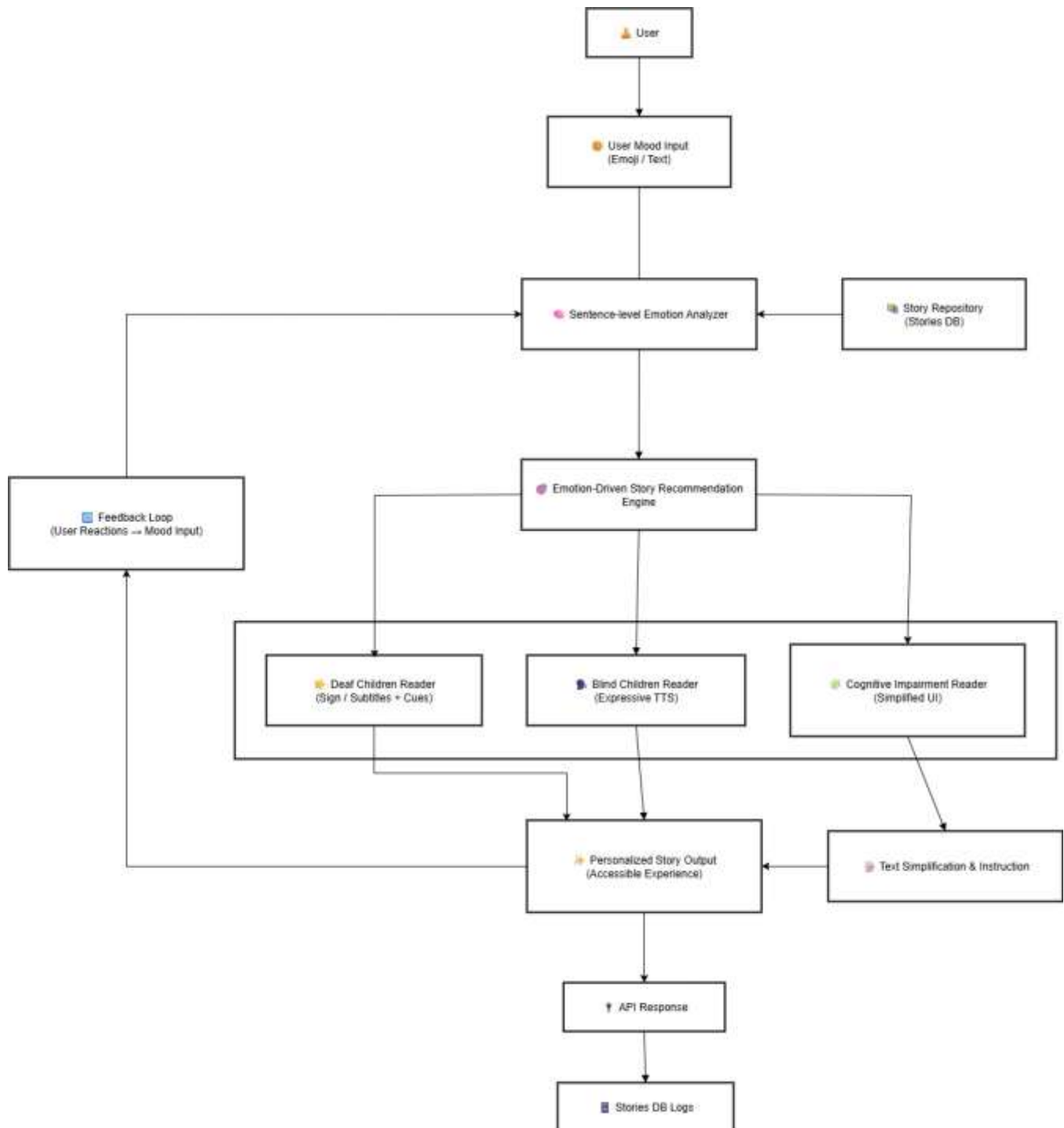


Figure 1: Component Diagram

1. User Interaction & Input

- The system starts with a **user** (child or facilitator) providing mood input through emojis or text.
- This ensures the system personalizes story recommendations based on **real-time emotional context**.

2. Story Repository (Knowledge Base)

- A centralized **Stories DB** stores diverse story content tagged with metadata such as themes, emotional tone, and difficulty level.
- This database serves as the foundation for personalized recommendations.

3. Emotion Analysis Layer

- The **Sentence-level Emotion Analyzer** interprets both user mood input and story metadata.
- It detects emotional states like happiness, sadness, curiosity, or anxiety, ensuring recommendations align with the child's emotional needs.

4. Recommendation Engine

- The **Emotion-Driven Story Recommendation Engine** acts as the core decision maker.
- It maps user mood to appropriate story content, balancing engagement, comfort, and accessibility.

5. Accessibility Pathways

- The system adapts story delivery for different child categories:
 - **Blind Children** → Expressive **Text-to-Speech (TTS)** for immersive narration.
 - **Deaf Children** → **Sign language or subtitles with cues** for inclusivity.
 - **Children with Cognitive Impairment** → **Simplified UI and instructions**, reducing cognitive load.

6. Content Simplification & Instruction

- An additional **text simplification and instruction module** ensures content remains understandable and emotionally appropriate for cognitive impairment scenarios.

7. Personalized Story Output

- All processing converges into a **personalized, accessible story output**, ensuring children experience content tailored to both their **mood** and **abilities**.

8. API Response & Database Logs

- The system provides **API responses** to external applications for scalability (integration with e-learning platforms, apps, etc.).
- **Stories DB Logs** record user activity and system decisions for future improvement.

9. Feedback Loop

- A **feedback mechanism** captures user reactions and feeds them back into the mood input.
- This creates a **continuous improvement cycle**, refining personalization over time.

3.4 Proposed Project Plan

- **Phase 1: Requirement Analysis and Ethics Approval**
Gather detailed functional and non-functional requirements. Obtain ethical clearance for working with children with disabilities.
- **Phase 2: Data Collection and Framework Development**
Develop tools for multimodal emotion data collection (emoji, text, voice). Build and train the multimodal emotion detection framework.
- **Phase 3: Story Tagging and Recommendation Engine Development**
Implement sentence-level emotional tagging algorithms. Develop the adaptive story recommendation engine integrated with the emotion detection system.
- **Phase 4: System Integration and Initial Testing**
Integrate all components into a unified system. Conduct initial internal testing for functionality and usability.
- **Phase 5: User Trials and Evaluation**
Conduct user studies with children with disabilities to assess accuracy, engagement, and therapeutic effects.
- **Phase 6: Refinement and Documentation**
Analyze feedback and refine system performance. Complete documentation and prepare reports for publication and deployment.

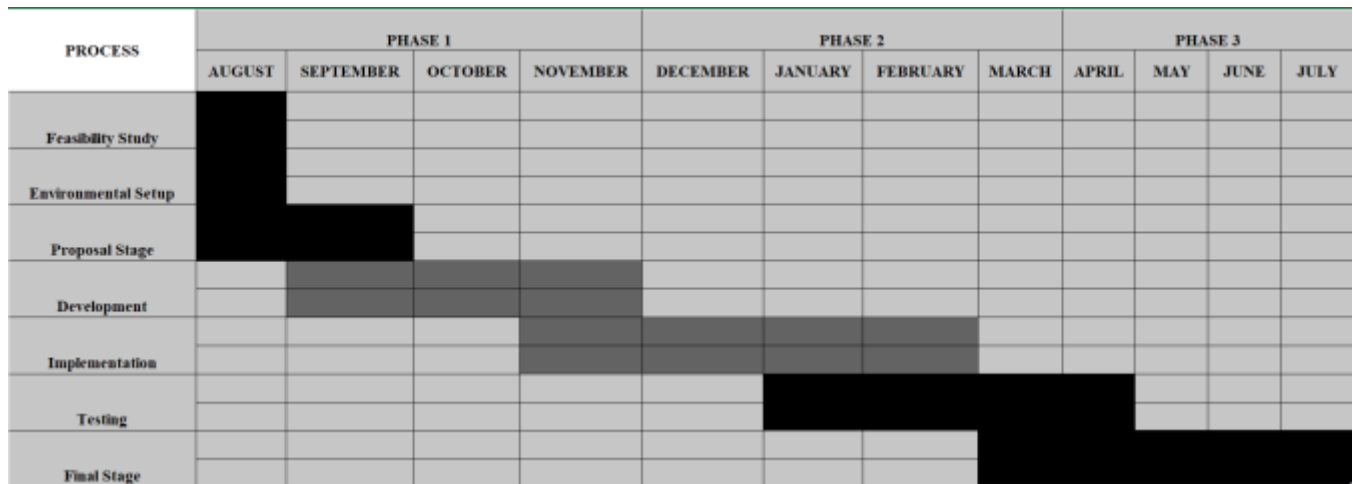


Figure 2: Gantt Chart

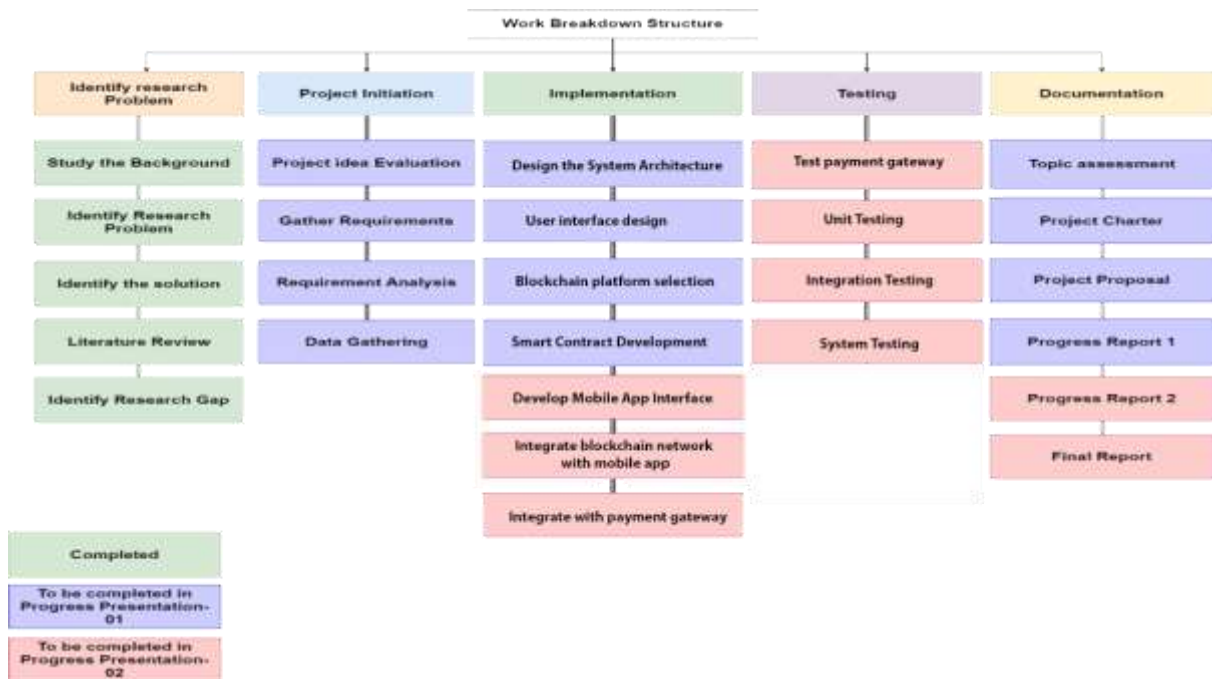


Figure 3: Work Breakdown Structure

3.5 Anticipated Conclusion

The proposed Emotion-Driven Story Recommendation System is expected to enhance the emotional engagement of children with disabilities by accurately detecting their emotional states through accessible multimodal inputs such as emoji selection, text sentiment, and voice emotion recognition. The sentence-level emotional tagging of stories will enable a tailored, mood-congruent storytelling experience that supports emotional regulation and well-being.

Results:

Preliminary implementation anticipates improved accuracy in emotion detection and greater user engagement compared to existing unimodal systems, with effective alignment of story content to detected moods.

Application:

The system could be widely applied in special education, therapeutic contexts, and adaptive learning environments, promoting inclusivity and supporting emotional development among children with diverse disabilities.

Real-World

Use:

In practical settings, educators and therapists can leverage this tool to personalize storytelling, track emotional progress, and enhance interactive learning experiences for children.

Enhanced

User

Engagement:

By dynamically adapting narratives to the child's current emotional state, the system fosters deeper emotional connections, improved motivation, and soothing therapeutic benefits.

3.6 Commercializatio

1. The target markets for the proposed emotion-driven storytelling system primarily include special education centers, therapy clinics, and online adaptive learning platforms where personalized emotional support is essential. Offering the system as a cloud-based service allows easy access and scalability, with customizable content libraries to cater to the diverse needs of users across regions and cultures.
2. Collaborations with educational technology companies and healthcare providers will be pursued to facilitate widespread adoption and ongoing enhancement of content and features. Additionally, the technology holds commercial potential through licensing agreements or integration into existing assistive technology products, enabling broader use and continuous innovation.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The system shall collect emotional inputs from users via emojis, text, and voice.
- It shall perform multimodal emotion detection by analyzing emoji selections, text sentiment, and voice tone.
- The system shall tag story sentences with emotional labels and recommend story segments matching the user's current emotion.
- Real-time adaptive storytelling with dynamic story content adjustment based on detected mood is required.

4.2 User Requirements

- The interface must be accessible and user-friendly for children with various disabilities, including cognitive and sensory impairments.
- Input methods should accommodate different communication abilities, offering options beyond text and voice.
- The system shall protect user privacy and provide clear consent mechanisms for data collection.
- Users require engaging, culturally relevant, and emotionally supportive stories.

4.3 System Requirements

- Hardware requirements include a fast processor, adequate memory (minimum 8GB RAM), microphone and speakers for audio input/output.
- Software must support NLP, speech processing, sentiment analysis, and machine learning algorithms.
- The system should be compatible with major operating systems and support cloud deployment for scalability.
- Network connectivity is essential for content updates and remote processing.

4.4 Non-Functional Requirements

- Performance:

The system must process multimodal emotional inputs and adapt story content in real-time with minimal latency to ensure a smooth, engaging user experience. Algorithms should be optimized for quick detection and recommendation without perceptible delay.

- Scalability:

The architecture must support scaling to accommodate multiple users concurrently, supporting expansion to large user bases without degradation in response time or accuracy. Cloud infrastructure and distributed computing techniques should be leveraged.

- Security:

User data, especially sensitive emotional and personal information, must be securely stored and transmitted using encryption. Access controls and anonymization techniques should be implemented to protect user privacy and comply with relevant data protection regulations.

- Usability:

The system interface must be intuitive, accessible, and usable by children with different disabilities. It should provide multiple interaction options (emoji, text, voice) ensuring inclusivity and ease of emotional expression.

- Reliability:

The platform must be robust, offering consistent operation with minimal downtime. Fault-tolerant design, error handling mechanisms, and backup procedures should ensure continuous availability and data integrity.

5. BUDGET AND BUDGET JUSTIFICATION

Component	Amount (LKR)
Web Hosting	9,600
Development Tools and Techniques	22,400
Cloud Hosting and Data Storage	12,000
Paper publishing cost	50,000
Total	94 000

Table 2: Budget Distribution

6. REFERENCES

- [1] T. Vijayakumaran, T. Thavachelvam, A. Gnaeswaran, T. Sandanayake, K. Sumathipala, and S. Bandara, “Development of Digital Storytelling Platform for Children based on Emotions,” in *Proc. 6th Int. Conf. Information Technology Research (ICITR)*, Moratuwa, Sri Lanka, Dec. 2021, pp. 1–6, doi: 10.1109/ICITR54349.2021.9657431.
- [2] S. Lankadari and B. Rodrigo, “Storytelling for Preschool Children in Sri Lanka: Social and Emotional Development Between Linear and Non-linear Storytelling Approaches,” in *Advances in Design and Digital Communication II*, Jan. 2022, pp. 452–465, doi: 10.1007/978-3-030-89735-2_37.
- [3] A. Kengatharan *et al.*, “Emotional Storyteller for Vision Impaired and Hearing-Impaired Children,” *Int. J. Recent and Innovation Trends in Computing and Communication*, vol. 11, no. 10, pp. 670–677, 2023, doi: 10.17762/ijritcc.v11i10.8562.
- [4] R. Wilson, “The Integration of Advanced AI-Enabled Emotion Detection and Adaptive Learning Systems,” *Journal of Educational Technology*, vol. 57, no. 3, pp. 145–163, Mar. 2025.
- [5] S. Tan and Y. Zhu, “CatHill: Emotion-Based Interactive Storytelling Game for Cognitive Therapy,” in *Proc. ACM Digital Library*, 2023, doi: 10.1145/3544549.3585639.
- [6] N. Capuano, “Emotion-Based Digital Storytelling for Risk Education,” *Int. J. Comput. Eng. Educ.*, vol. 3, no. 2, pp. 120–134, 2014.
- [7] L. Brown, “Advanced Affective Computing Techniques for Real-Time Emotion Detection,” *IEEE Trans. Affect. Comput.*, vol. 12, no. 1, pp. 50–62, Jan. 2024.
- [8] A. Kumar and S. Das, “Multimodal Emotion Recognition for Interactive Learning Applications,” *Int. J. Hum.-Comput. Stud.*, vol. 130, pp. 33–45, Jul. 2024.
- [9] H. Nguyen, “Emotion-Based Adaptive Learning: A Comprehensive Overview,” *Front. Comput. Sci.*, vol. 11, 2024.

- [10] P. Lee and J. Harris, “Integrating Machine Learning for Enhanced Emotional Engagement in E-Learning,” *J. AI Res.*, vol. 67, pp. 89–105, 2023.
- [11] C. M. Wegemer, E. Halim, and J. Burke, “Designing an Adaptive Storytelling Platform to Promote Civic Education in Politically Polarized Learning Environments,” *arXiv preprint* arXiv:2507.00161, Jun. 2025. [Online]. Available: <http://arxiv.org/pdf/2507.00161.pdf>
- [12] K. Lee and M. Wang, “Emotion-Driven Adaptive Storytelling,” *InTheValley.blog*, Aug. 2024. [Online]. Available: <https://inthevalley.blog/ai-generated-articles/industry-insights/game-tech/emotion-driven-adaptive-storytelling/>
- [13] M. Chen *et al.*, “Research on Classroom Emotion Recognition Algorithm Based on Deep Learning,” *PMC*, Aug. 2022. [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9377850/>

